

APPLIED PROBABILITY AND STATISTICS



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AI-Based Smart Campus Simulation System

A probability-based dashboard for modeling campus operations
and generating practical operational insights.

Final Term Project Report

Prepared for presentation and project defense

Generated from the implemented simulation app

1. Executive Summary

The AI-Based Smart Campus Simulation System is an interactive web application designed to model common campus operations using probability distributions. The system allows users to change assumptions, run simulations, view summary statistics, inspect charts, and read operational insights. It demonstrates how probability and statistics can support decision-making in a university environment.

Project focus: converting theoretical distributions into a practical dashboard for student arrivals, cafeteria queues, WiFi failures, shuttle waits, parking occupancy, classroom occupancy, and energy consumption.

Main Objectives

- Apply probability distributions to realistic campus operations.
- Build an interactive simulation dashboard using Python, HTML, CSS, and JavaScript.
- Generate useful statistics including average, minimum, maximum, and AI-style insights.
- Present charts that make simulated behavior easier to compare and explain.
- Support final project presentation with a clear, visual, and defensible system design.

2. Problem Statement

Campus managers often need to estimate demand and risk before making decisions. For example, they may need to know whether student arrival pressure is high, whether cafeteria queues are acceptable, whether WiFi failures are manageable, or whether energy use is above the planning baseline. Real-world campus data changes constantly, so the project uses simulation to explore likely outcomes under different assumptions.

Research Question

How can probability distributions be used to simulate campus operations and communicate operational insights through a simple web-based dashboard?

3. Methodology: Probability Distributions

Each dashboard tab is mapped to a distribution that matches the behavior of the campus process. Counts use count-based distributions, waiting times use continuous distributions, and occupancy or failure cases use success/failure models.

Campus Component	Distribution	Reason for Selection
Student Arrivals	Poisson Distribution	Counts arrivals per minute, a typical event-counting problem.
Cafeteria Queues	Exponential Distribution	Models time between service completions or waiting/service durations.
WiFi Failures	Binomial Distribution	Models repeated success/failure outcomes across devices.
Shuttle Bus Waiting Times	Uniform Distribution	Models waits within a known minimum and maximum range.
Parking Occupancy	Binomial Distribution	Models whether each parking space is occupied or available.
Classroom Occupancy	Normal Distribution	Models attendance clustering around an average.
Campus Energy Consumption	Normal Distribution	Models daily usage fluctuating around a mean baseline.

4. System Architecture

User Inputs	Validation	Distribution Sampling	Summary + Insight	Dashboard Chart
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The application uses a lightweight Python HTTP server. The backend provides configuration data and simulation results through API endpoints. The frontend renders tabs, inputs, metrics, insights, dynamic backgrounds, and a canvas chart. This simple architecture keeps the project easy to run, explain, and demonstrate.

Layer	Technology	Responsibility
Backend	Python HTTP server	Defines components, validates inputs, samples distributions, returns JSON.
Frontend	HTML, CSS, JavaScript	Displays dashboard, handles interactions, draws charts, updates visuals.
Visualization	Canvas API and CSS	Shows bar charts, metric cards, animated logo, and rotating campus visuals.

5. Dashboard Modules

Module	User Inputs	Output and Insight
Student Arrivals	Mean arrivals / minute, Minutes to simulate	Counts student arrivals per minute.
Cafeteria Queues	Average service time (min), Students served	Models waiting and service times.
WiFi Failures	Network devices, Failure probability, Simulation rounds	Models success/failure outcomes for access points.
Shuttle Bus Waiting Times	Minimum wait (min), Maximum wait (min), Passengers	Models waits inside a fixed minimum and maximum range.
Parking Occupancy	Parking spaces, Occupancy probability, Observation rounds	Each parking space is occupied or available.
Classroom Occupancy	Mean students, Standard deviation, Room capacity, Classes	Attendance tends to cluster around an average.
Campus Energy Consumption	Mean kWh / day, Standard deviation, Days	Daily usage fluctuates around a mean value.

6. Sample Simulation Results

The following table shows one deterministic demonstration run using default assumptions. Actual values can change when a user changes the inputs or reruns the simulation.

Component	Distribution	Sample Average	Generated Insight
Student Arrivals	Poisson Distribution	31.15 students/min	Arrival pressure is moderate. Add temporary entry guidance if the average rises above 45 students per minute.
Cafeteria Queues	Exponential Distribution	3.09 minutes	Service speed is acceptable for normal demand.
WiFi Failures	Binomial Distribution	1.9 failures/round	Failure behavior is within a manageable range.
Shuttle Bus Waiting Times	Uniform Distribution	9.8 minutes	Average shuttle waiting time is comfortable.
Parking Occupancy	Binomial Distribution	373.89 occupied spaces	Parking availability should remain manageable.
Classroom Occupancy	Normal Distribution	36.05 students	Classroom utilization is balanced.
Campus Energy Consumption	Normal Distribution	1512.59 kWh/day	Energy use is close to the planning baseline.

7. User Interface and Interaction Design

The interface was designed to be presentation-friendly and easy to scan. The homepage introduces the project with the Kyungdong University logo and rotating scenario visuals. The dashboard uses top navigation cards for the seven simulations, blue selected states, fixed navigation, metric cards, and a fitted graph surface.

Key UI Features

- Fixed top navigation for easy movement between Home and Dashboard.
- Seven simulation cards placed at the top of the dashboard for quick switching.
- Dark/light blue selected state for the active simulation card.
- Dynamic scenario visuals that describe campus activities while keeping the logo visible.
- Responsive grid behavior for smaller screen sizes.
- Chart surface that resizes to fit the available dashboard area.

8. Testing and Validation

The project was validated by checking the JavaScript syntax, confirming the local server serves updated files, and testing the dashboard behavior through the browser. Inputs are clamped to safe minimum and maximum values in the backend before simulations are calculated, reducing the chance of invalid or extreme data breaking the dashboard.

Validation Area	What Was Checked	Result
Backend input handling	Numeric values are parsed and clamped to configured limits.	Invalid entries fall back safely.
Frontend behavior	Tabs, run button, metrics, charts, and backgrounds update correctly.	Interactive workflow works locally.
Visual design	Navbar, buttons, logo, chart panel, and dashboard cards were refined.	Presentation-ready layout.

9. Limitations and Future Improvements

- The current system uses simulated data rather than live campus sensor or administrative data.
- AI insights are rule-based text responses rather than machine-learned predictions.
- Charts are currently drawn as bar/histogram views; future work could add line charts, comparisons, and export tools.
- A database could be added to store historical simulation runs and compare scenarios over time.
- User authentication and admin roles could be added for a production campus platform.

10. Conclusion

This project successfully demonstrates how applied probability and statistics can support campus operation planning. By connecting distributions to real campus problems, the system turns mathematical concepts into an interactive tool that can be explained, tested, and extended. The final application is suitable for demonstrating simulation, user input handling, statistical summaries, visual analytics, and practical decision support.